

Seeing Machines

A Multimodal Archive Companion - Semester Final Project

Course	Program	Instructors	Submission deadline
CompSci for Designers 2	MA Design for Digital Futures TH Nürnberg, Fakultät Design	Moritz Schwind Christopher Kopic	July 9, 2026 23:59h

1 Project Overview

Over the semester you have built a retrieval-augmented generation (RAG) pipeline over text: tokenization and chunking, text embeddings, similarity-based retrieval, sampling parameters, and grounded generation with lightweight language models. In this final project you extend that pipeline into the visual domain using vision-language models (VLMs) and multimodal embeddings.

You will curate a personal image corpus — a collection you know intimately — and build a working system that lets you search, question, and converse with it. Because you know your own archive, you will immediately notice when the machine sees something that is not there. Documenting and explaining these moments of machine mis-seeing is as important as the working prototype.

Your corpus

50–150 images (up to 300 for Level 3). Examples: your own photography, a build log of a personal project, scans of a zine or sketchbook archive, screenshots from a game, a documented collection. Use only images you have the rights to use; if other people are identifiable in your images, obtain their consent or choose a different corpus.

Choosing your level

The project is offered at three levels that build strictly on one another: every level reuses the previous level's system as a component. Choose the level that matches your current technical comfort — **your level choice does not cap your grade** (see rubric, Section 5). If you finish your level early, you continue upward on your own project rather than starting over.

2 The Three Levels

L1 The Finder for students with little or no prior AI/coding experience

What you build: a semantic image search engine over your archive. A joint text–image embedding model (CLIP or SigLIP) embeds all images once; your text query is embedded into the same space; cosine similarity returns the top-k images in a simple Gradio interface. No captioning, no text generation — pure retrieval. A starter notebook with the embedding loop is provided; your work is corpus curation, query experimentation, tuning top_k and similarity thresholds, and the interface.

Deliverables:

- Working search prototype (Colab notebook + Gradio interface).
- A short corpus statement: what the collection is, why you chose it, how you curated it.
- A **retrieval atlas**: at least 10 documented queries — successes and failures — each with a brief explanation grounded in embedding-space reasoning (why did this query return these images?).

Escalation hook: your embedded corpus and retrieval function carry into Level 2 unchanged.

L2**The Companion**

core brief, for students comfortable with the semester's notebooks

What you build: full multimodal RAG. A VLM (Gemma 3 4B, quantized) generates a structured description of every image — **you design the description schema and the prompt that produces it**; this is the central design decision of the level. The descriptions are embedded as text, retrieval runs over them, and the language model generates grounded answers while the retrieved images are displayed alongside. Caption once and cache the results to Google Drive; never re-caption per session.

Built-in comparison: keep the Level 1 CLIP retrieval as a baseline and compare both routes on the same queries. When does joint-embedding retrieval beat caption-mediated retrieval, and when is it the other way around?

Deliverables (in addition to Level 1):

- Chat-with-your-archive prototype (retrieval + grounded generation + images shown in the answer).
- Description schema and prompt, with a written justification of the fields and vocabulary you chose.
- Side-by-side comparison of the two retrieval routes on at least 5 shared queries.
- A **mis-seeing dossier**: documented hallucinations, counting failures, spatial confusions, or skewed descriptions the VLM produced about your corpus, with your analysis of each case.

Escalation hook: you now own two independent retrieval routes over one corpus — exactly the raw material Level 3 needs.

L3**The Critic**

for advanced students

What you build: two additions on top of Level 2. First, **hybrid retrieval**: fuse CLIP similarity and caption-text similarity into a weighted score, with the weighting exposed in the interface or justified in writing. Second, **true multimodal grounding**: at answer time, pass the retrieved images themselves (maximum 3–4) into the VLM's context instead of answering from captions alone. Small VLMs degrade noticeably with multiple images in context — documenting that degradation is a legitimate finding, not a failure.

Evaluation layer: construct a gold-standard query set (20–30 queries with known-correct images from your corpus) and measure precision@k across all three retrieval routes: CLIP-only, caption-only, and hybrid.

Deliverables (in addition to Level 2):

- Hybrid retrieval system with justified or user-adjustable fusion weighting.
- Multimodal answering mode (retrieved images in the VLM context), with observations on quality and degradation.
- Gold-standard query set and a short **evaluation report**: precision@k for all three routes, with numbers and an interpretation of *why* the routes differ.

3 Technical Constraints & Colab Guidance

All three levels run on the **free Google Colab tier** (T4 GPU). Follow these rules to avoid the common traps:

- **Develop on a CPU runtime; attach the GPU only for captioning and generation runs.** Free-tier GPU quota is dynamic and opaque — idling with a GPU attached can lock you out near the deadline. Level 1 runs fine on CPU end to end.
- **Use the pinned setup from the course notebook.** The T4 does not support bfloat16, and Gemma 3 has known float16 instability on this card; use the provided 4-bit quantized loading path (recent transformers + bitsandbytes, Unsloth, or a GGUF build) rather than assembling your own stack.
- **Caption once, cache to Drive (mandatory).** Captioning ~150 images takes 20–45 minutes on a T4; write the results to a JSON file in your Google Drive so a disconnect costs you nothing.
- **Do not switch to Gemma 3 1B to save memory** — the 1B variant is text-only; vision starts at 4B.
- **Level 3: cap retrieved images at 3–4.** Each image costs the model roughly 256 tokens, and answer quality drops as image count rises.
- **Fallback:** if you exhaust Colab's GPU quota, Kaggle's free tier offers roughly 30 GPU-hours per week with more predictable availability.

- **Interface expectation: functional Gradio, no more.** Spend your hours on the pipeline and the documentation, not on frontend polish.

4 Timeline & Checkpoints

Session	Date	What happens / what you must have ready
VLM lecture	June 17, 2026	Brief released. Before the next session: choose your level, assemble your corpus, and confirm you can run the starter notebook.
Support session 1	_____	Checkpoint: your level's retrieval working end to end (query in, images out). Bring failures — this session exists to unblock you.
Support session 2	_____	Checkpoint: the full loop plus interface running. Remaining runway goes to documentation and (Level 3) evaluation.
Submission & demo	July 9, 2026 23:59h	All deliverables submitted (Section 6); short live demo of your prototype.

5 Grading Rubric

Grading is **criteria-based within your chosen level**: an excellent Finder can outscore a sloppy Critic. If you escalate beyond your level, you are assessed against the criteria of the highest level you complete coherently — a half-finished escalation does not replace a finished lower level.

Criterion	Weight	What excellent looks like
Technical implementation	30%	The pipeline works end to end and you can explain every stage in the vocabulary of the course: how your images or captions are tokenized and embedded, what similarity is computed, what top_k and sampling parameters do, and where generation is grounded. Cached, reproducible, no dead cells.
Design decisions & justification	25%	Choices are deliberate and defended: corpus curation, description schema and prompt (L2/L3), parameter settings, fusion weighting (L3), and interface decisions. The justifications show judgment, not defaults.
Critical reflection	25%	The retrieval atlas / mis-seeing dossier / evaluation report goes beyond description: failures are analyzed mechanistically (why did the model see this?), and limitations are stated honestly. Depth of analysis is weighted heavily at every level.
Documentation, demo & communication	20%	A reader who was not in the course could understand what the system does, how it works, and what it cannot do. The demo runs; the writing is clear; visuals support the argument.

6 Submission Requirements

- **Colab notebook(s)**, shared with view access, that run top to bottom on a fresh runtime (with cached captions loaded from the included files).
- **Cached artifacts**: your embeddings and (L2/L3) caption JSON, so grading does not require re-running GPU jobs.
- **Project documentation (GitHub Pages)**: a published, publicly accessible GitHub Pages site documenting your project in the style of a research project page. Two models: dreambooth.github.io (the canonical project-page format) and ricardomehl.github.io/publication/LatentWorld_Documentation (a student project from this program at realistic scale, including process documentation, code snippets, honest dead ends, and personal reflection). Expected sections: title, your name, chosen level, course and semester; a teaser visual with a one-line tagline; a short abstract; motivation and corpus statement; approach (with a pipeline diagram); a results gallery with captioned retrieval examples *and* failure cases (your retrieval atlas / mis-seeing dossier / evaluation report lives here); a short project reflection; a brief limitations section; sources and references. Note: GitHub Pages on free accounts requires a public repository, so publish only images you are allowed to share publicly.
- **PDF export of the documentation page**: a PDF of your published site (browser print-to-PDF is fine), submitted together with the link, so an archivable version of your documentation exists independently of the live site.
- **Corpus access**: the image set itself or, if it cannot be shared, a representative sample of at least 20 images plus a description of the full set.
- **Live demo** (approx. 5 minutes) in the final session: one working query path, one documented failure, one thing you would build next.

A note on honesty: documented failure analyzed well is worth more in this course than concealed failure or inflated claims. Your archive, your machine, your judgment.